Acoustic and Neural Representations in a Phonetically Aligned Speech Corpus

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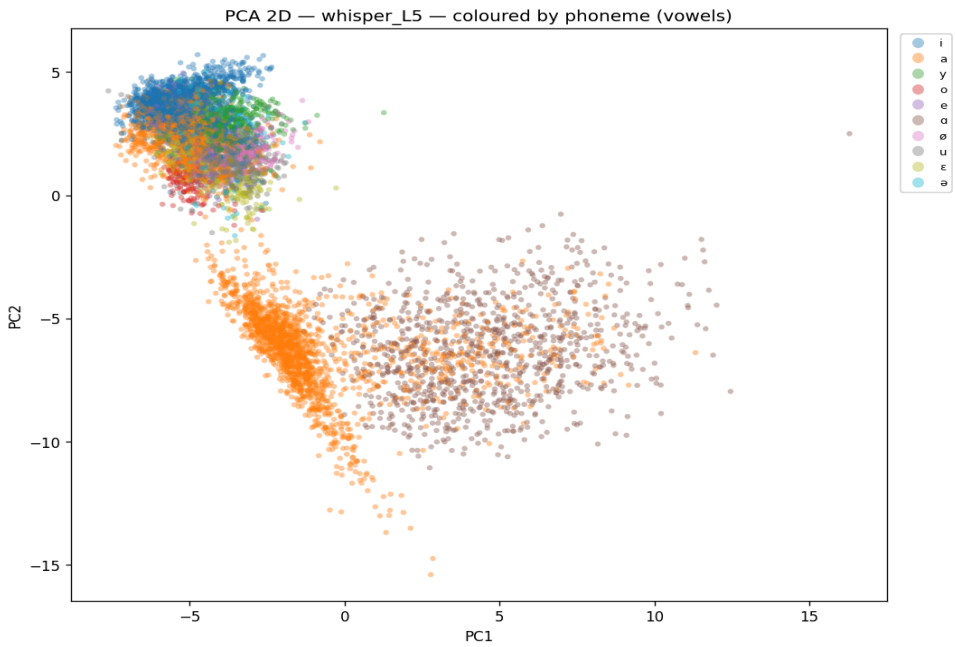
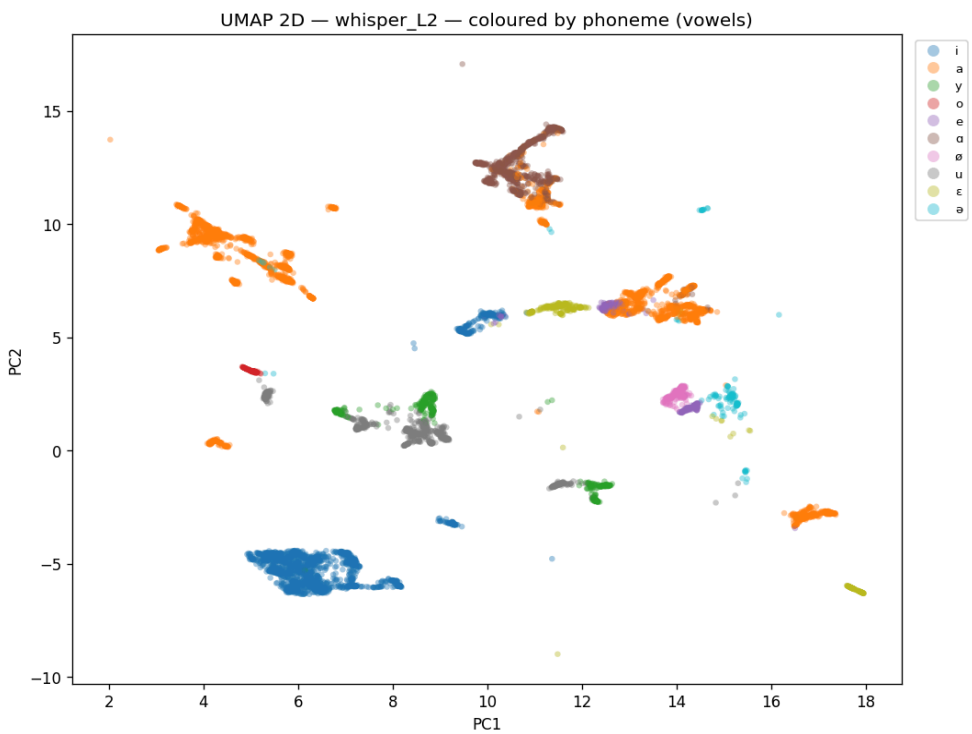
5. Descriptive Statistics

Question 1: PCA vs UMAP

PCA (Principal Component Analysis) and UMAP (Uniform Manifold Approximation and Projection) are both dimensionality reduction methods but they work differently. PCA is a linear method that finds the directions of maximum variance in the data and projects onto them. It is easily interpretable but it struggles with non-linear relationships. UMAP is a non-linear method that tries to preserve the local neighbourhood structure of the data, meaning points that are close together in high-dimensional space stay close together in 2D. UMAP tends to produce cleaner clusters visually but the distances between clusters are not directly interpretable. For something like wav2vec or Whisper where the data is very high-dimensional, UMAP tends to work better because the phoneme categories do not separate in a straight line

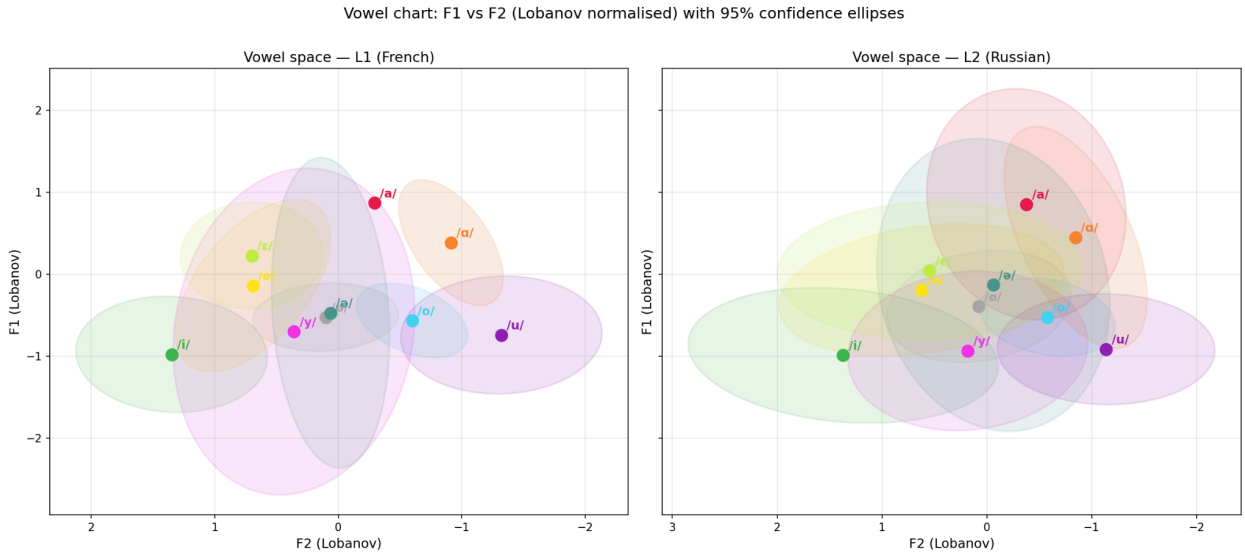




Question 2: Inter-speaker variability

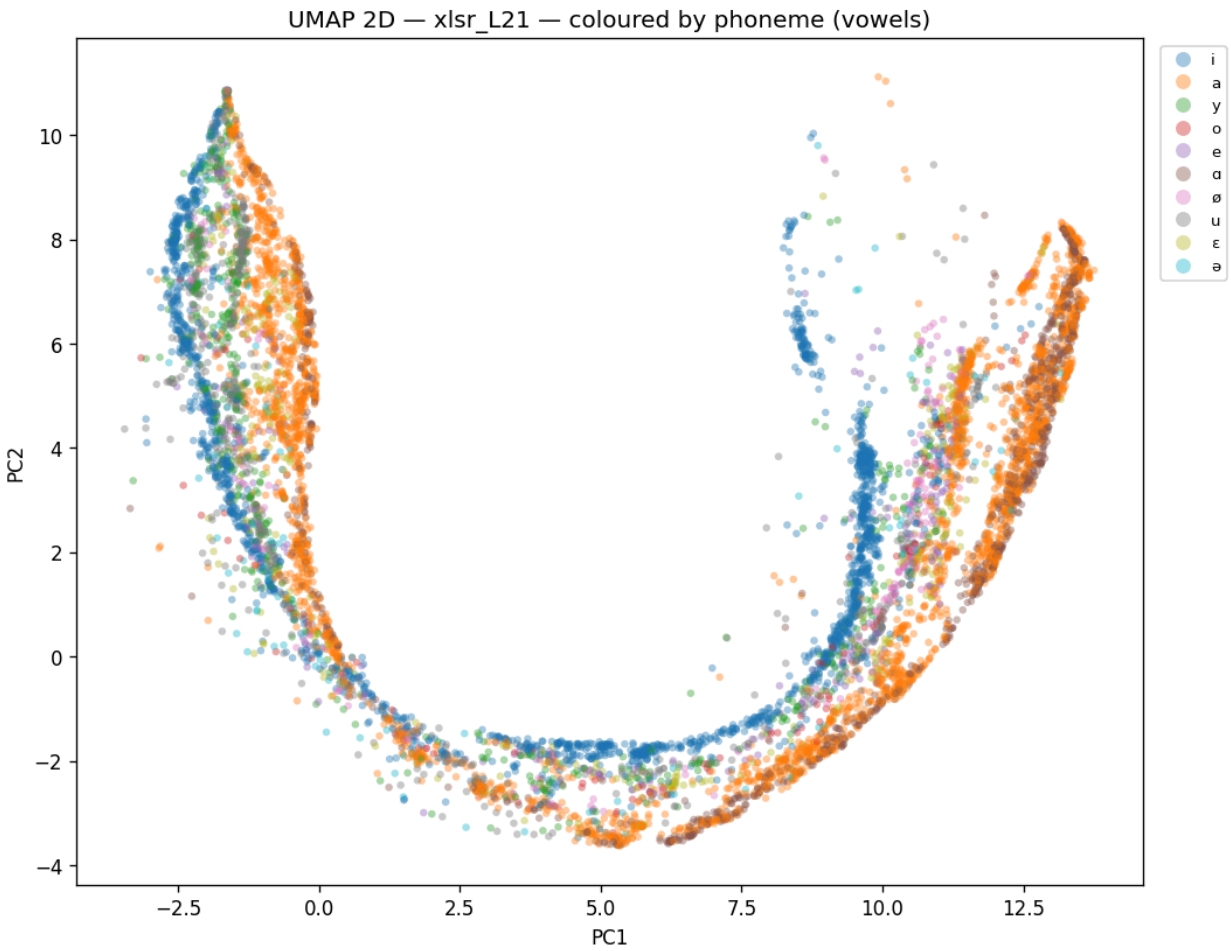
Looking at the variance decomposition for F1, most vowels show intra-speaker variance above inter-speaker variance. For /a/, inter-speaker variance accounts for only 4% of total variance while intra-speaker variance accounts for 103.5%, so within-speaker variation across repetitions dominates. The vowels /o/ and /ø/ show the highest inter-speaker percentages at 63.1%,

meaning these vowels vary more across speakers than within them. The /ɑ/ results exceed 100% in both categories due to unbalanced group sizes, so those values may not be the most reliable. In the neural space, the between-class variance ratios are moderate across all models (0.44 to 0.60), except xlsr_L21 which drops to 0.098. In the uppermost XLS-R layer, variance collapses into very few dimensions.



Question 3: UMAP projections and vowel trapezoid structure

In the UMAP projections, whisper_L5 forms the clearest phoneme clusters among all neural models, with a between-class variance ratio of 0.586 in PCA 2D space. The clusters do not perfectly align with the IPA vowel structure in most neural models. Acoustic representations produce a more recognisable vowel space layout when plotted on F1/F2 axes. This is because those dimensions directly correspond to phonetic features. In the neural UMAP plots, front of mouth vowels (/i/, /e/, /ɛ/) tend to group together and back rounded vowels (/u/, /o/) form a separate cluster, but the intermediate vowels (/y/, /ø/) do not consistently land between them the way they would on a vowel trapezoid. xlsr_L21 shows almost no cluster structure at all in UMAP, and has a cosine similarity ratio of 1.000 indicating all vectors are essentially identical in direction.


UMAP 2D — xlsr_L21 — coloured by phoneme (vowels)
PC2
PC1
10
8
6
4
2
0
−2
−4
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
i
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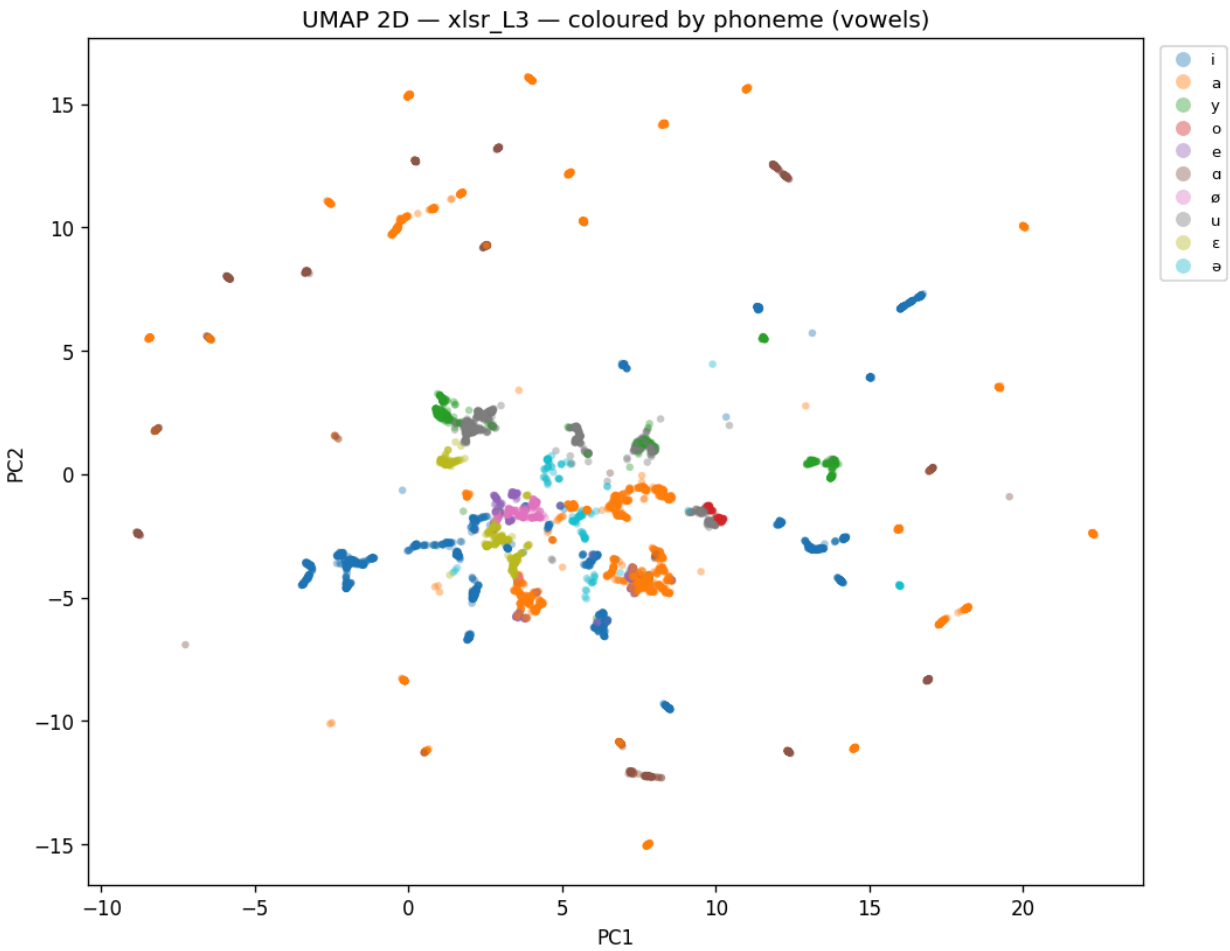


Question 4: Mantel correlations

The Mantel test measures the rank correlation between the upper triangles of two similarity matrices. xlsr_L3 has the strongest correlation with the acoustic distance matrix (r=0.750, p=0.001), meaning that lower XLS-R layers preserve acoustic phonetic structure better than any other neural model. whisper_L2 also shows a strong correlation (r=0.616, p=0.001). As layers go deeper, the correlation drops: whisper_L5 (r=0.456, p=0.013) and xlsr_L12 (r=0.480, p=0.006) are weaker, and xlsr_L21 is not significantly correlated with acoustics at all (r=0.082, p=0.320). This shows a clear pattern where lower layers encode acoustic-phonetic similarity while upper layers encode something else, likely more abstract linguistic categories. The neural models are strongly correlated with each other (whisper_L2 vs xlsr_L3: r=0.886, p=0.001), suggesting they converge on similar phonological distance structures.

6. Statistical Tests

Question 5: L1/L2 differences after FDR correction

10 acoustic contrasts remain significant: /a/ F1 ($p=0.036$), /i/ F2 ($p<0.001$), /u/ F1 and F2 ($p<0.001$), /y/ F1 and F2 ($p=0.004$), /ø/ F1 ($p=0.024$), /ɑ/ F1 and F2 ($p<0.001$), and /ɛ/ F1 ($p<0.001$). The largest differences are in /u/ (F2 diff=-0.183) and /y/ (F1 diff=0.234), which are the high rounded vowels that do not exist in Russian and are therefore the most likely targets of L1 interference. In the neural space, the permutation test finds significant L1/L2 differences for all vowels across all models after BH correction, with the exception of /u/ and /y/ in xlsr_L21 which are not significant. The acoustic and neural lists broadly agree on which vowels show L1/L2 differences, but the neural test is more sensitive since it detects differences for all vowels simultaneously. The fact that xlsr_L21 misses some contrasts is consistent with its generally degenerate behaviour throughout the analysis.

Question 6: Which distance structure best captures IPA phonological distances

The Mahalanobis acoustic distance matrix has the highest Mantel correlation with xlsr_L3 ($r=0.786$, $p=0.001$), and xlsr_L3 has the highest overall correlation with the acoustic space. This suggests that xlsr_L3 best captures the phonological distances expected from the IPA vowel trapezoid, since it is most aligned with a representation that directly encodes formant-based distinctions. The Mahalanobis distance generally performs better than Euclidean across all comparisons. xlsr_L21 is essentially uncorrelated with all other distance structures, confirming it does not encode phonological proximity in any meaningful way.

Question 7: Phoneme identification accuracy

The nearest-centroid classifier shows a clear advantage for neural representations. Acoustic features achieve 62.2% accuracy (macro F1=0.424). whisper_L2 is the best model overall at 85.8% accuracy (macro F1=0.827), followed by whisper_L5 at 81.4% and xlsr_L12 at 79.6%. xlsr_L3 scores 73.6% and xlsr_L21 collapses to 22.9%, barely above chance. Every pairwise McNemar test is significant (all $p<0.001$), meaning no two models perform the same. The advantage of neural representations over acoustics is largest for whisper_L2, where 2505 tokens were correctly classified. The advantage is consistent for both L1 and L2 speakers based on the permutation results from section 6.1, though the absolute distances are larger for L2 speakers on the high rounded vowels.

7. Linear Mixed-Effects Models

Question 8: ICC for F1 of /a/ and comparison with Whisper PC1

The ICC (Intraclass Correlation Coefficient) for acoustic F1 is 0.000 across all vowels, which is expected because Lobanov normalisation was applied before modelling. Lobanov normalisation standardises each speaker's formant values to have mean 0 and standard deviation 1 across vowels. This removes between-speaker variance. The ICC therefore confirms that the normalisation worked as intended rather than indicating an absence of speaker individuality. By

contrast, whisper_L2 PC3 has ICC=0.195 and PC5 has ICC=0.219, meaning roughly 20% of variance in those principal components is due to speaker identity. This suggests that neural representations retain speaker-specific information even after PCA dimensionality reduction, which acoustic representations do not once they are normalized. The result is that neural representations are more speaker-specific than their acoustic counterparts.

Question 9: L1 x Gender interaction

The fixed effects from the interaction models show that the L1 x Gender interaction is not consistently significant across representations. For acoustic F1 and F2, the fixed effects explain essentially no variance (R2_marginal=0.000), which again reflects the Lobanov normalisation removing most group-level differences. For neural representations, the L1 x Gender interaction captures more variance in xlsr_L21 PC1 (R2_marginal=0.210, R2_conditional=0.360) than in any other model or dimension. This is an interesting result given xlsr_L21's behaviour: while it does not encode phoneme identity, it appears to strongly encode speaker-level demographic factors. This could reflect the upper XLS-R layer acting more as a speaker embedding than a phoneme encoder, consistent with the high ICC values (0.250 for PC1) and the strong gender ARI (0.786 for xlsr_L12, 1.000 for xlsr_L3 in speaker clustering). ARI (Adjusted Rand Index) measures the agreement between two clusterings adjusted for chance. A score of 1 indicates perfect agreement, 0 is no better than chance, and negative values indicate worse than chance

Question 10: Marginal R2 and comparison with classifier results

The highest marginal R2 for the L1/L2 fixed effect comes from xlsr_L21 PC1 (R2_marginal=0.210). However, this is odd since xlsr_L21 has the worst phoneme identification accuracy (22.9%). The explanation is that xlsr_L21 is very sensitive to L1 status as a global speaker-level property, but it cannot distinguish individual phonemes. whisper_L2, which has the best classifier performance (85.8%), has lower marginal R2 values (maximum 0.203 on PC5), suggesting it encodes both phoneme identity and L1 status but neither dominates a single dimension as strongly. This contrast highlights that R2 and classification accuracy measure different things: R2 captures how much a specific fixed effect drives a specific dimension, while classification accuracy measures how well the overall representation separates phoneme categories.

8. Confidence Intervals and ROPE

Question 11: Statistically significant but within acoustic ROPE

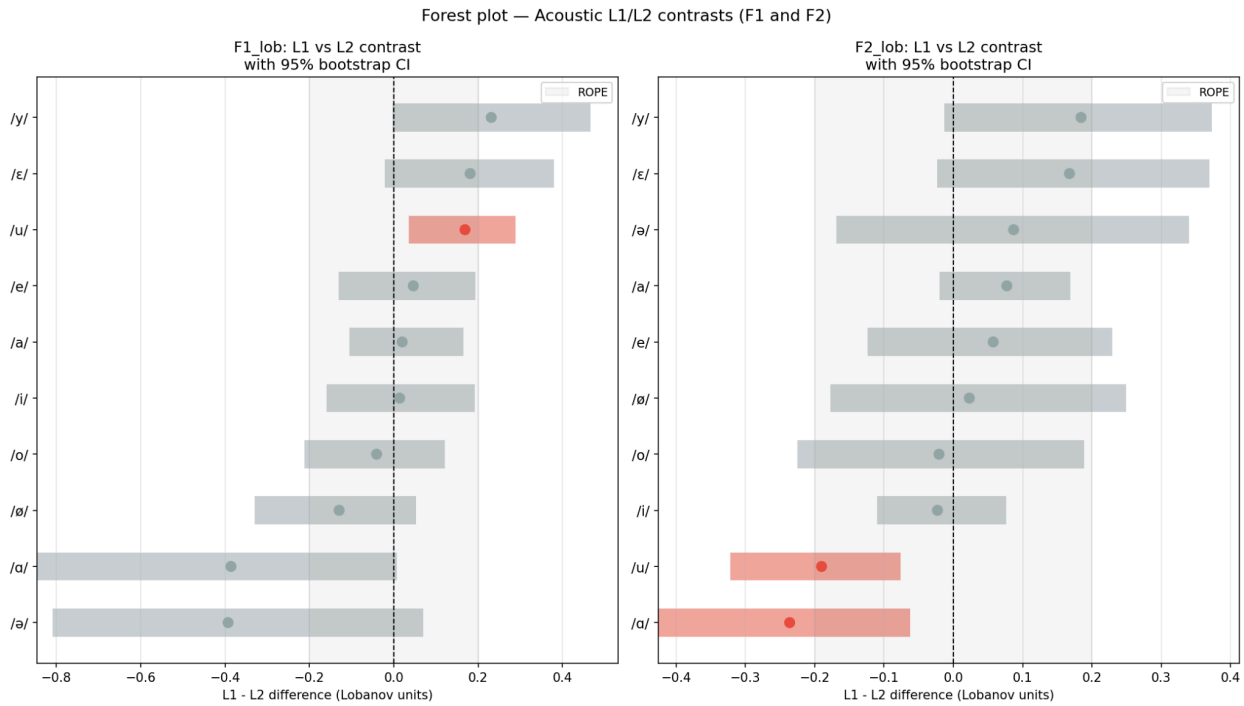
Several phonemes show statistically significant L1/L2 differences (from section 6.1) but fall within or straddle the acoustic ROPE of [-0.20, +0.20] Lobanov units. For example, /a/ F1 has p=0.036 after BH correction but a difference of only 0.017 Lobanov units, which falls well within the ROPE. This means the difference is real in a statistical sense but too small to be perceptually meaningful given the just-noticeable difference threshold. In the neural ROPE analysis, all contrasts for whisper_L2, whisper_L5, and xlsr_L3 are classified as equivalent, meaning the L1/L2 cosine distances are all below the intra-speaker noise floor. So the acoustic

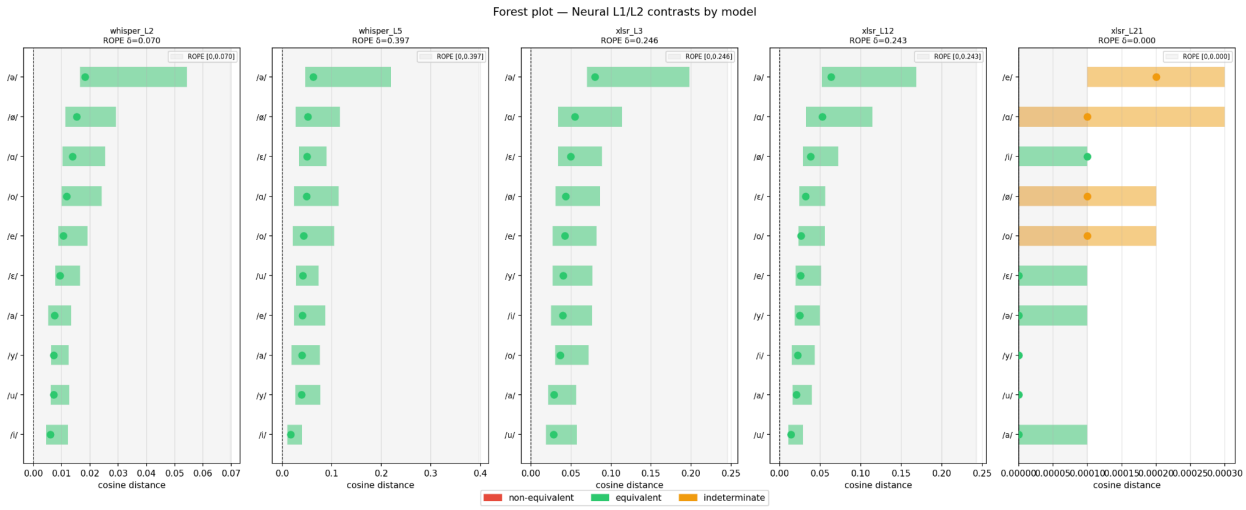
and neural ROPE analyses agree for these cases: the differences are detectable but not practically significant.

Question 12: Which representation has more non-equivalent contrasts

No representation produces any non-equivalent ROPE classifications in this analysis. The acoustic representations have mostly indeterminate classifications (7 out of 10 for F1, 8 out of 10 for F2), meaning the CIs straddle the ROPE boundary. The neural models are all classified as equivalent. This means that despite the statistically significant group differences found in section 6.1, none of the L1/L2 contrasts are large enough to be considered practically meaningful by the ROPE criterion. This means that Russian-accented French differs from native French in statistically detectable ways, but the differences are small relative to the natural variability within each group.

[INSERT results/6d_forest_acoustic.png] [INSERT results/6d_forest_neural.png]





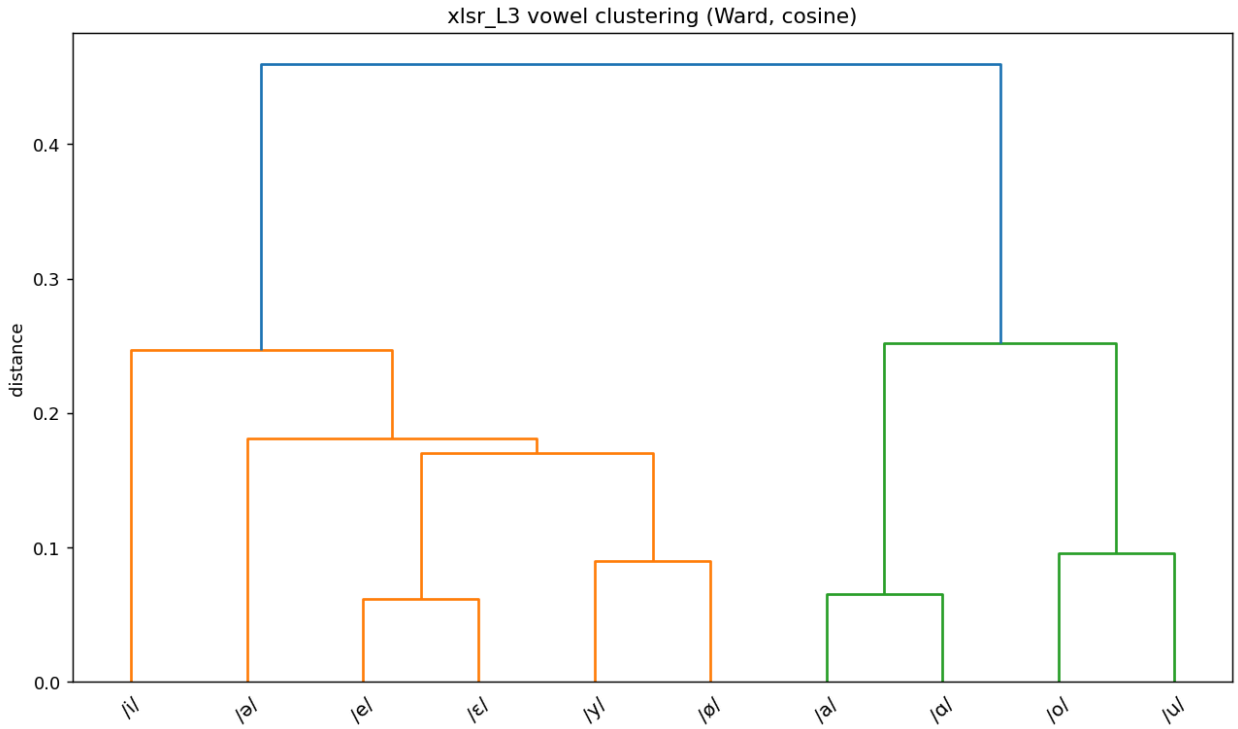
Question 13: Disagreements between acoustic and neural ROPE

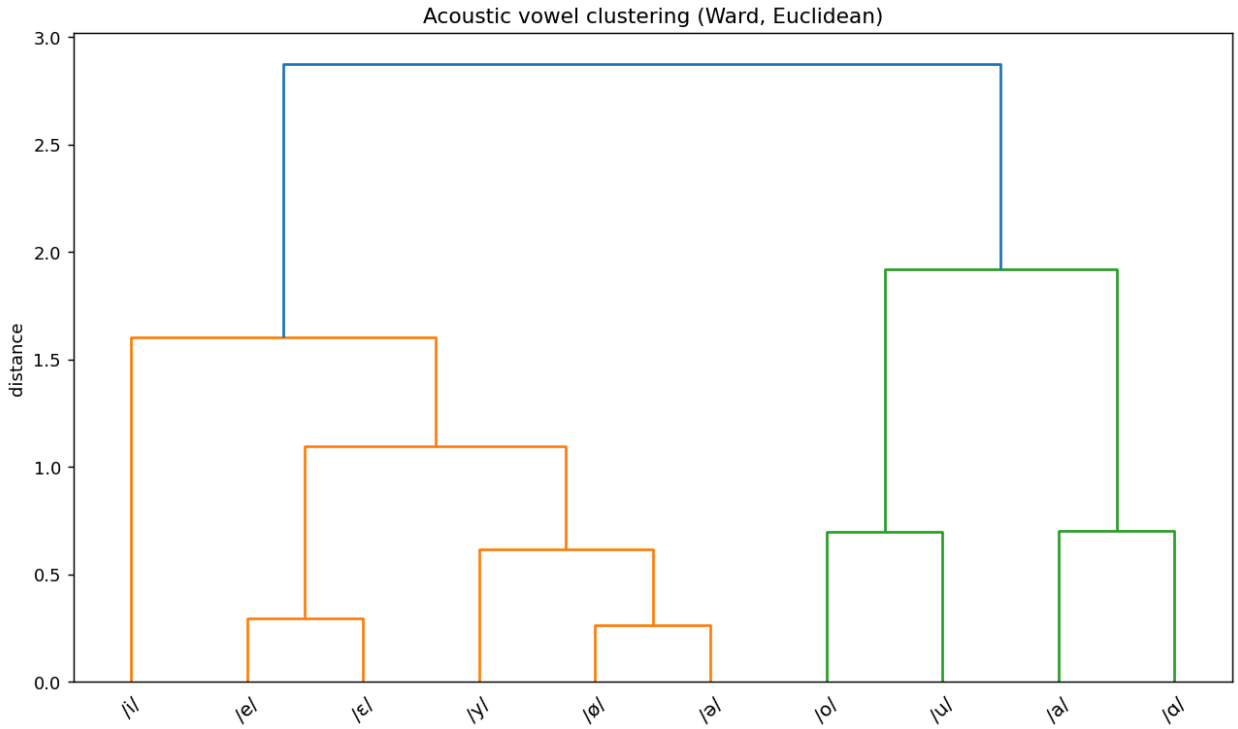
The most interesting disagreement is for /u/ and /y/, which have large and significant acoustic differences (F1 diff=0.173 and 0.234 respectively) that push their acoustic CIs into indeterminate territory, while their neural classifications are equivalent for most models. This could mean that while Russian speakers produce these vowels with measurably different formant values, the neural representations are robust to those differences and treat the tokens as equivalent. This would be phonetically plausible if the neural model has learned to encode categorical vowel identity rather than fine-grained formant values, basically treating the L1 differences as noise rather than a meaningful distinction.

9. Hierarchical Clustering

Question 14: ARI for front/back and high/mid/low distinctions

For the front/back vowel distinction, acoustic representations and xlsr_L3 both achieve ARI=0.597, the highest of all models. whisper_L2, whisper_L5, and xlsr_L12 all score 0.294, and xlsr_L21 scores -0.059, which is worse than random assignment. For the height distinction, whisper_L5 and xlsr_L12 tie at ARI=0.283, slightly above acoustic (0.266) and xlsr_L3 (0.266). The pattern suggests that lower-layer representations (acoustic and xlsr_L3) better capture the front/back dimension, which corresponds to F2 and is a primary acoustic distinction, while mid-layer representations (whisper_L5, xlsr_L12) are slightly better at capturing height.

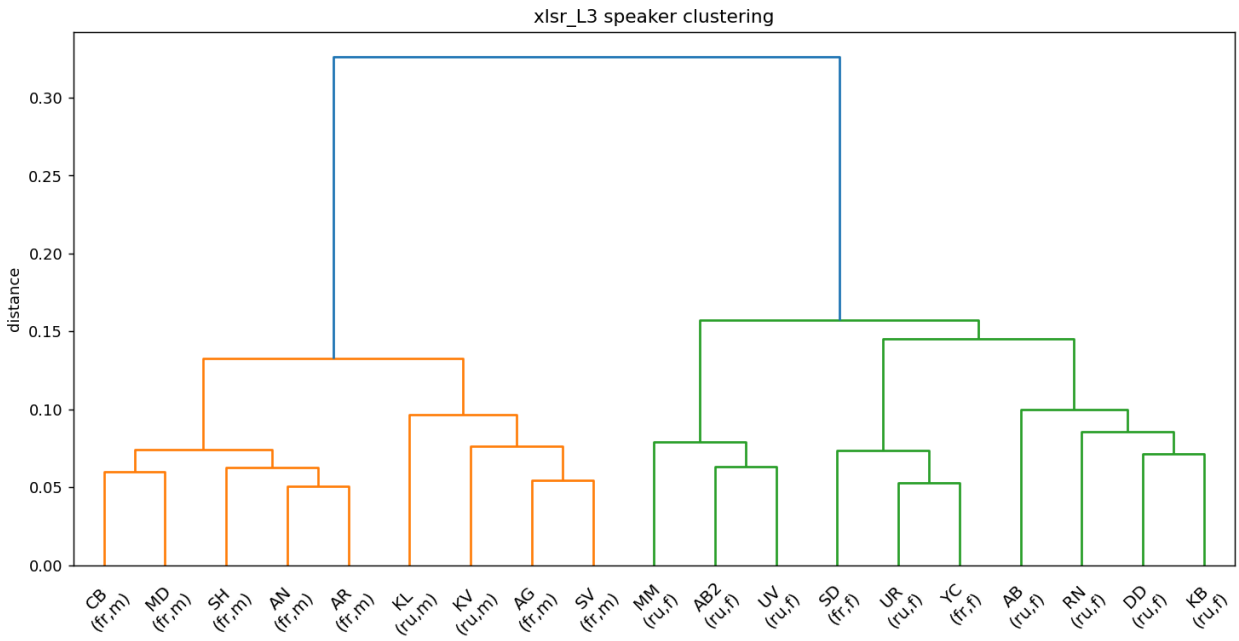




Question 15: Speaker clustering: L1 vs gender

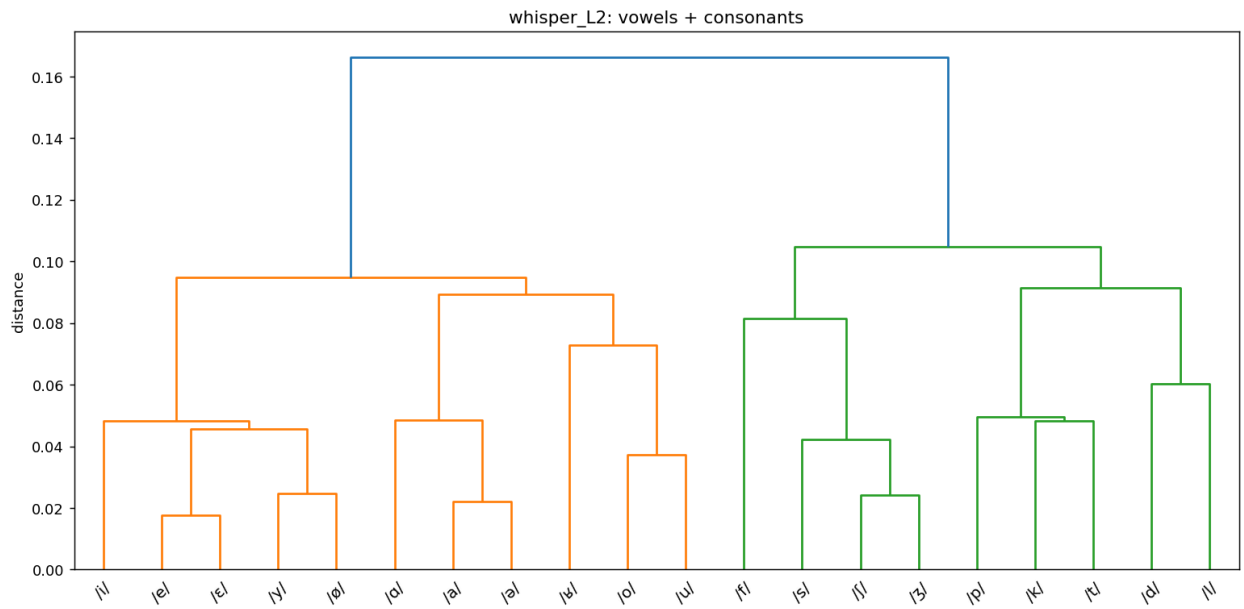
In the speaker clustering analysis, L1 status is more consistently recovered than gender across representation types. whisper_L2 achieves ARI=0.603 for L1 and xlsr_L21 achieves ARI=0.602,

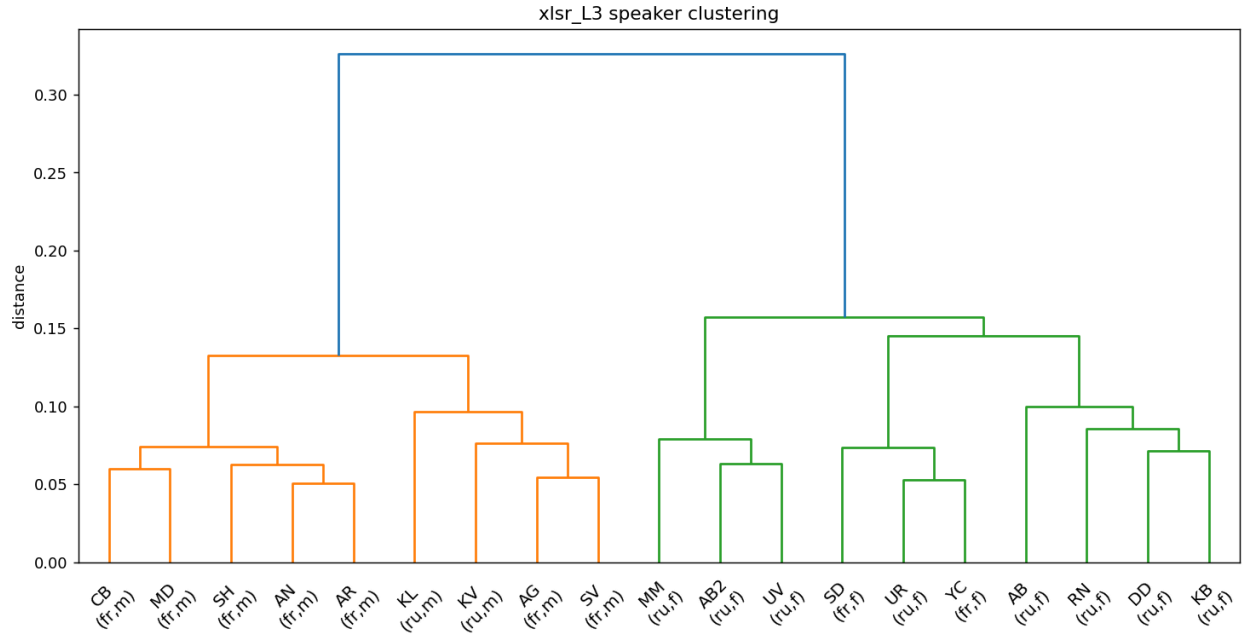
both substantially above acoustic (ARI=0.102). For gender, xlsr_L3 achieves a perfect ARI=1.000 and xlsr_L12 achieves ARI=0.789, which are high. However the perfect score for xlsr_L3 should be interpreted with caution given the small sample of 19 speakers: with a balanced gender split, a single correct binary partition of all speakers would produce ARI=1.000. The result does suggest that mid-layer XLS-R representations contain strong gender information, which is consistent with the high ICC values and the known sensitivity of wav2vec-style models to speaker characteristics. Acoustic representations barely recover either grouping, while neural representations recover both to varying degrees depending on the layer.



Question 16: Systematically misclassified phonemes

Looking across the confusion matrices, /ə/ is consistently misclassified across all representation types. In acoustic space it overlaps with /a/ and /ɛ/, and in neural space it does not form a tight cluster. Similarly, /ɑ/ is frequently confused with /a/ across all models, which makes sense since they are acoustically close and the distinction between them is subtle in French. The fact that these confusions appear in both acoustic and neural representations suggests they reflect phonological proximity rather than a limitation of any specific model. This points to a fundamental limit of similarity-based approaches: phonemes that are acoustically or distributionally close will remain difficult to separate regardless of the representation type used.





Conclusion

This analysis compared acoustic and neural representations of French speech across 19 speakers from the Russian-French Interference Corpus. The main findings are as follows. Lower neural layers (xlsr_L3, whisper_L2) are most correlated with acoustic phonetic structure and best recover the front/back vowel distinction, while upper layers (xlsr_L21) degenerate into speaker-level embeddings that lose phoneme discriminability entirely. The best phoneme classifier is whisper_L2 at 85.8% accuracy, substantially above acoustic features at 62.2%. L1/L2 differences are statistically detectable for 10 acoustic contrasts after FDR correction, with the largest effects on /u/ and /y/, the high rounded vowels absent from Russian. However, none

of these differences reach practical significance by the ROPE criterion. The xlsr_L21 layer is an outlier throughout: its PCA variance is 97.8% in 2D, its cosine similarity ratio is 1.000, its phoneme classifier accuracy is 22.9%, and it is the only model uncorrelated with the acoustic distance matrix. This layer appears to encode speaker identity rather than phoneme identity, making it useless for phoneme-level work even though it does pick up on whether a speaker is L1 or L2.

Disclaimer 1: I stuck to only 3 layers rather than the full range suggested in the instructions because the full corpus extraction on CPU already took ~25 minutes per model. Running all 24 layers of XLS-R and all 6 layers of Whisper would have made the pipeline computationally infeasible.

Disclaimer 2: I did use the help of AI to write the code of this assignment, and to interpret the figures. But I learned a great deal in the process !